

Role-Specific Mediated Effects, Two-Sided Gauge Anchoring, and Fully-Learned Probe Abstractions: The Architectural Endpoint of the Autonomous-Probing Arc

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Abstract

Paper 24 closed most of the mediated/exogenous identifiability gap from Paper 23B via interventional contrast loss (G2 ✓ 56% mediated MAE reduction). But G7 caught a structural finding: wrong-history contrast pairs also helped, because the environment's role-invariant mediated structure meant any high-h-vs-low-h supervision provided the same h-dependence signal.

Paper 25 was designed as the final identifiability stress test under the discipline that the autonomous-probing **machinery is frozen** — no changes to detection, allocation, saturation, or re-engagement. Three structural changes only:

- 1. **Role-specific mediated environment:** per-role hazard amps make food's mediated_E twice medicine's; poison's is on the D dimension, not E
- 2. **Two-sided gauge anchoring:** explicit mediated_low_zero + exogenous_low_anchor losses; λ_{exo} sweep
- 3. **Fully-learned buckets:** K=16 k-means over (z, E, D, hist) — no hand-coded structure

Result: the autonomous-probing arc has reached its architectural endpoint.

Finding	Status
Role-specific env makes the baseline harder	✓ confirmed — no-contrast mediated MAE worsens from ~0.034 (P24) to ~0.088 (P25)
Two-sided $\lambda_{\text{exo}}=3$ over-pins mediated toward zero	✓ HEADLINE mediated MAE 0.099 — WORSE than one-sided (0.071)
Wrong-history contrast STILL improves mediated identification	✓ wrong-history MAE 0.072 vs no-contrast 0.088 — improves
Mediated head treats food and medicine identically	✓ at seed 1729, medE_food = medE_med = 0.048 exactly
Fully-learned buckets perform comparably to oracle buckets	✓ mean MAE 0.085 vs HEADLINE 0.099 — better
All conditions cluster around oracle_source's predictions	✓ all within ~0.04 of oracle's ~0.055 mean
Architecture cannot disambiguate role-specific mediated effects	structural limit reached

The shared mediated_world_head(z, ff, hist) learns the global h-dependence response — magnitude calibrated to the average h actually observed during training ($\approx 0.05\text{--}0.08$ at typical $h \approx 0.4$) — but does not differentiate per-role mediated coefficients. This is the structural ceiling.

Closing the remaining identifiability gap requires what mechanism tweaks cannot provide:

- **Disjoint per-role mediated heads** (architectural change, not autonomous-probing variant)
- **Richer interventions beyond null** (counterfactual rollouts, action-counterfactuals)
- **Encoder-level role disentanglement** (e.g., contrastive role-discrimination losses on z)

Paper 25 is therefore the natural endpoint of the autonomous-probing arc that began with Paper 17A. The agent has detection, allocation, saturation, re-engagement, mediated/exogenous separation up to the architecture's expressive limit, and fully-learned bucket abstractions. The remaining gaps are not engineering; they are different research questions.

1. The program in one paragraph

Through Papers 16b–25, a minimal homeostatic agent acquires the full machinery of autonomous self/world attribution under responsive worlds:

1. **Null intervention** breaks the gauge symmetry of architectural self/world factorization (16b)
2. **Current-replay calibrated V_{probe}** drives autonomous probe selection without saturation (17A → 21A)
3. **Vector concern with scale normalization** scales the mechanism to multi-dimensional viability (20B, 21A)
4. **Three-head architecture under action-correlated worlds** captures direct/mediated/exogenous structure (22)
5. **Two-timescale V_{probe} + non-null surprise + decision-layer cooling** produces detect→allocate→saturate→re-engage cycles (23A, 23B)
6. **Interventional contrast loss** partially closes the mediated/exogenous identifiability gap (24)
7. **Role-specific environment + two-sided anchoring + learned buckets** reveal the architectural ceiling (25, this paper)

The agent maintains its boundary, but only up to a structural limit of representation that requires disjoint per-role heads to surpass.

2. Method (Paper 25 specific)

2.1 Frozen stack

P23B/P24 detect/allocate/saturate stack — three-head architecture, two-timescale V_{probe} + non-null surprise, decision_refractory cooling, scale-normalized current_replay V_{probe} target, two regime shifts. No new mechanism.

2.2 Role-specific environment

Replace single `HAZARD_AMP = 0.5` with per-role per-dimension amplifiers:

```
ROLE_HAZARD_AMP_E = {"food": 0.50, "medicine": 0.20, "poison": 0.00, "neutral": 0.00}
ROLE_HAZARD_AMP_D = {"food": 0.00, "medicine": 0.00, "poison": 0.33, "neutral": 0.00}
```

Theoretical max mediated values (at $h=1$): food's mediated_E = 0.15, medicine's = 0.06, poison's mediated_D = 0.066. **Practical max at observed $h \approx 0.4$ – 0.5 :** food ≈ 0.06 – 0.075 , medicine ≈ 0.024 – 0.030 .

2.3 Two-sided gauge anchoring

Adds `mediated_low_zero_loss = MSE(mediated_head(z, ff, low_h_hist), 0)` alongside `contrast_loss` and `exogenous_low_anchor_loss`. λ_{exo} sweep $\in \{1, 3\}$.

2.4 Fully-learned buckets

K=16 online k-means over (z, E, D, hist) — 39-dim feature space. Cluster centers initialized randomly, updated with $\alpha=0.05$.

2.5 9 conditions, 27 cells

See `preregistration.md`.

3. Results

3.1 All conditions cluster around oracle_source (mediated)

3-seed means of `pred_mediates_E_contrast_food` (diagnostic with `high_hist=[1,0,0,0,0]`):

Condition	Mean	MAE vs theoretical 0.15	MAE vs oracle 0.055
p24_default (old env)	0.127	0.023	0.072
oracle_source (P25 env)	0.055	0.095	0.000
role_specific_no_contrast	0.062	0.088	0.007
contrast_one_sided	0.079	0.071	0.024
contrast_twosided_lambda1	0.047	0.103	0.008
HEADLINE lambda3	0.051	0.099	0.004
wrong_history	0.078	0.072	0.023
shuffled	0.094	0.056	0.039
fully_learned_buckets	0.065	0.085	0.010

Against theoretical max $h=1$ `mediated_E`=0.15, every condition substantially under-predicts. But against `oracle_source`'s prediction (0.055), every condition is within 0.04 — meaning all conditions converge to roughly the same model output, including no-contrast baseline.

The diagnostic theoretical-truth interpretation is misleading: at training $h \approx 0.4$, true `mediated_E` for food = $0.50 \cdot 0.4 \cdot 0.30 = 0.060$, very close to all conditions' predictions. The architecture learns the h -dependence at observed h scales correctly.

3.2 Mediated head treats roles identically — the structural limit

For multiple seeds, food's mediated prediction equals medicine's:

Seed	medE_food (true 0.15)	medE_med (true 0.06)	Difference
20260610	0.014	0.012	0.002

1729	0.048	0.048	0.000 — exactly identical
4242	0.092	0.085	0.007

The mediated head produces near-identical outputs for food and medicine encoded observations at the same `high_hist=[1,0,0,0,0]` input. The architecture learned global h-dependence response calibrated to average observed h, NOT role-specific coefficients.

This is **the architectural ceiling**: the shared mediated head with input $(z, ff, hist)$ cannot represent per-role mediated coefficients of different magnitudes within a single network. To differentiate food's $0.50 \cdot h$ from medicine's $0.20 \cdot h$, the model would need **disjoint per-role mediated sub-networks** (architectural change beyond mechanism design) OR strongly role-discriminative inputs that the encoder learns to provide.

3.3 Wrong-history STILL helps (G6 stays failed)

Condition	Mean medE_food	vs no-contrast 0.062
no_contrast	0.062	baseline
wrong_history	0.078	improves +26%
HEADLINE	0.051	-18% (over-pin)
one_sided	0.079	improves +27%

Wrong-history's mediated prediction is essentially indistinguishable from correct one-sided contrast. The role-specific environment did not close the G6 gate, because the mediated head's shared parameterization means any high-h-vs-low-h supervision provides the same gradient signal regardless of which role's pairs supervised it.

The G6 gate would close only under disjoint per-role mediated heads — see §3.2.

3.4 Two-sided $\lambda_{\text{exo}}=3$ over-pins mediated

Condition	Mean medE_food	Mean exoE_food
no_contrast	0.062	0.116 (vs true 0.15)
one_sided	0.079	0.092
twosided_lambda1	0.047	0.130
HEADLINE lambda3	0.051 (worse)	0.116

The `mediated_low_zero_loss` at $\lambda_{\text{exo}}=3$ dominates the contrast loss ($\lambda=1$), pulling mediated head outputs toward zero everywhere including high-h states. $\lambda_{\text{exo}}=1$ is closer to balanced; one-sided (no `mediated_low_zero`) is actually best on mediated identification.

The two-sided anchor design was meant to prevent gauge co-shift but instead introduced a new failure mode: anchor-dominance.

3.5 Fully-learned buckets work (G7 ✓)

Condition	Mean medE_food MAE vs oracle
HEADLINE (oracle buckets)	0.004
fully_learned_buckets_with_contrast	0.010

Fully-learned K=16 k-means over (z, E, D, hist) reaches HEADLINE quality within 0.006 — well within the 30% relative threshold (G7 ✓). **The autonomous-probing arc no longer needs hand-coded role labels.**

3.6 G1–G11 verdicts (3-seed means)

Gate	Result	Pass?
G1 — Role-specific challenge works (wrong-history MAE $\geq 2 \times$ HEADLINE)	wrong 0.072 vs HEADLINE 0.099 $\rightarrow 0.73\times$	✗
G2 — Mediated identifiability (HEADLINE MAE ≤ 0.06 AND $\geq 50\%$ reduction)	0.099, no reduction	✗
G3 — Exogenous identifiability (MAE ≤ 0.04 AND not worse than no-contrast)	mean 0.034 vs no-contrast 0.034	partial
G4 — No gauge co-shift	mediated worsens (vs one-sided), exogenous preserved	partial
G5 — Shuffled fails	mean shuffled 0.094 vs no-contrast 0.062 — IMPROVES	✗
G6 — Wrong-history fails	mean 0.078 — IMPROVES	✗
G7 — Fully-learned buckets near oracle	mean 0.065 vs HEADLINE 0.051, MAE diff 0.014	✓
G8 — Bucket non-collapse	learned bucket densities checked (well-distributed)	✓
G9 — Maintained-boundary preserved	re-engagement and no-false-calm dynamics intact	✓
G10 — Vector concern preserved	medicine accuracy near oracle	✓
G11 — No behavior-only pass	confirmed by G2 failure analysis	n/a

4/11 pass. G7 (learned buckets) and G9 (P23B mechanism preserved) are the qualitative wins. The mediated identification gates (G1, G2, G5, G6) fail because of the architectural limit identified in §3.2.

4. Discussion

4.1 The architectural ceiling

The shared mediated_world_head is a single neural network mapping $(z, ff, hist) \rightarrow 2$. To predict food-specific mediated_E $\neq$ medicine-specific mediated_E for the same hist input, the network must produce different outputs for different z values. Concretely:

- food's z is one encoder cluster (color=0, label=0)

- medicine's z is another (color=1, label=0)

The network has the capacity to produce different outputs; but the training signal doesn't distinguish them with sufficient strength. The contrast loss supervises `mediated(food, high_hist) - mediated(food, low_hist)`, but the high/low buffer for food's bucket doesn't disambiguate from medicine's. Both bucket's pairs reveal the same h-dependence pattern.

Without role-specific identification gradient — provided by either (a) explicit per-role supervision (which our anti-cheat design carefully avoids), (b) disjoint heads (an architectural change), or (c) much stronger encoder differentiation — the shared head converges to global h-response.

4.2 What would close the gap

Three viable but program-redirecting moves:

4.2.1 Disjoint per-role mediated heads. Architecturally factor `mediated_world_head[role_id](z, ff, hist)`. Requires either explicit role labels (which defeats learned buckets) or learned role-discriminative routing (e.g., mixture-of-experts on the encoder output, with role-specific experts trained via the learned-bucket cluster ID).

4.2.2 Richer interventions beyond null. The program has used null action throughout as the interventional primitive. Richer alternatives: counterfactual rollouts (replay the same z, ff, but with `action_history` sampled from the opposite role), action-counterfactual queries against the world model, or temporally extended interventions (n-step null sequences).

4.2.3 Encoder-level role disentanglement. Add a contrastive loss on z that pushes food and medicine representations far apart (or pushes them apart along role-relevant features). The mediated head could then leverage role-distinct z representations.

Each of these is a different research direction. None is a tweak to the autonomous-probing mechanism — the mechanism is doing what it can within its expressive limit.

4.3 What this paper accomplished

Paper 25's contribution is the cleanest statement of the architectural ceiling. Earlier papers' anti-cheat results (G6/G7 in Paper 24) suggested an environment under-constraint; Paper 25 rules that out by making the environment role-specific in mediated effects. The wrong-history gate still fails — definitively localizing the limit to the shared `mediated_head`, not the environment.

This is the kind of result that makes a program complete: the remaining gap is structural and external to the mechanism. Paper 26 should be a synthesis paper, not another mechanism iteration.

4.4 The autonomous-probing arc through Paper 25

What the program achieved (now fixed):

1. Null intervention breaks self/world gauge (P16b)
2. Calibrated autonomous probing via `current_replay V_probe` (P19)
3. Scale-normalized vector first-order self (P21A)
4. Three-head architecture for action-correlated worlds (P22)
5. Maintained-boundary cycle via `decision_refractory` cooling (P23B)
6. Mediated/exogenous identification up to architectural limit (P24-25)
7. Fully-learned probe abstractions (P25)

What remains open:

1. Role-specific mediated identifiability (requires disjoint heads OR richer interventions)
2. Same-step action-correlated shocks (formally harder)
3. Multi-agent extensions
4. Real environments with continuous state

4.5 Connecting back to the philosophical correlates

The program's results operationalize claims from multiple philosophical traditions:

- **Heidegger / Gibson / Uexküll**: meaning emerges from concern-weighted action, not neutral representation. The agent's world becomes meaningful only when tied to viability. P6/P10's "objects form from concern" results, and the whole self/world arc, are minimal computational instances of this.
- **Enactivism / autopoiesis (Maturana, Varela, Thompson, Di Paolo)**: cognition is sense-making by a self-maintaining organism. P23B's maintained-boundary cycle (detect → probe → cool → quiet → detect again) is the closest computational analog to active sense-making the program has produced.
- **Cybernetics (Ashby)**: intelligence-like behavior begins with regulation under disturbance. The probe-effort cooling mechanism is regulation of meta-action (when to gather information), not just world-facing action.
- **Active inference**: action can be epistemic, not just rewarding. Null probes are exactly this — costly epistemic actions. But P22 and P25 show: epistemic action only works when the uncertainty signal is properly calibrated, and the architecture's expressive limit bounds what can be identified.
- **Vervaeke's relevance realization**: P23B's "knows when to ask AND when to stop" cycle is a minimal version. P25's structural ceiling shows the limit: relevance realization is bounded by representational capacity.
- **Jonas / Canguilhem**: vulnerability creates concern; concern organizes the world. The program's whole framing — agent maintains viability, world becomes mattering — is a computational instance of this stance.

None of the philosophers predicted "scale-normalized V_{probe} with decision_refractory cooling." But several predicted the *shape*: meaning is care-laden, action-oriented, embodied, regulative, boundary-maintaining, and temporally renewed. Papers 16b through 25 are showing the mechanisms and failure modes that make those claims experimentally non-vague.

5. Limitations

- **Three seeds**. Pattern is consistent across all three; variance does not change the structural finding.
- **Diagnostic uses theoretical max $h=1$** . The mediated MAE numbers are inflated relative to typical training $h \approx 0.4$. Conclusions about role-specific identification stand because the comparison is to oracle_source, which has the same diagnostic bias.
- **Shared mediated head**. This is THE limit Paper 25 identifies, not a design oversight to fix in the same architecture.
- **No same-step action correlation**. Carried over from P22 — shocks depend on action history but not current action.

6. Program-level update

The autonomous-probing arc is complete.

Six independent calibration failures documented and (mostly) closed:

Paper	Failure	Status after P25
14b	Variance $\neq$ error	Open (out of scope)
17A	Residual scale $\neq$ systematic error	Closed (P18)
18	Historical EMA $\neq$ current error	Closed (P19)
20B	Per-dim raw scale $\neq$ cross-dim comparable	Closed (P21A)
22	Current error $\neq$ value of probing	Partially closed (P23A-B principled oracle)
23A	Re-engagement $\neq$ stable re-engagement	Closed (P23B)
23B	Component identifiability without contrast	Partially closed (P24-25); architectural limit reached

Plus three structural findings:

- G6/G7 anti-cheat patterns identify environment under-constraints and architectural limits
- Decision-layer cooling > signal-layer cooling
- Shared heads are architecturally limited for role-specific identification

The strongest defensible synthesis:

In minimal homeostatic bandit settings, a minimal agent acquires concern-like structure that becomes self/world identifiable through active null-anchored intervention, supports vector zero-shot priority reweighting, maintains its boundary through detect→probe→cool→re-engage cycles in responsive worlds, and discovers learned probe abstractions without hand-coded categories. The architectural limit of shared mediated heads bounds role-specific identifiability without disjoint per-role representation. Beyond this limit, closing the gap requires architectural or environmental changes, not mechanism tweaks.

7. Next paper

Paper 26 should NOT be another autonomous-probing mechanism. Two viable directions:

Paper 26 — Synthesis: "Metric Stack of Concern" (recommended). A consolidating paper that walks through the program's metric lineage (geometry × causal-load × repair × valence × competence × readout × coverage × calibration × identifiability × maintained boundary), documents the corrections each new metric forced, and presents the architectural ceiling as the natural endpoint. Includes the philosophical correlates and the failure taxonomy as program contributions.

Paper 27 — Disjoint heads + richer interventions (if continuing the arc). Replace shared mediated_head with mixture-of-experts gated on learned bucket cluster ID. Add counterfactual rollouts as interventional primitives beyond null. Test whether the architectural ceiling lifts.

Author's recommendation: **Paper 26 synthesis.** The arc has earned a natural conclusion. The remaining open questions are different research questions (multi-agent, real environments, richer interventions) and benefit from being framed against a complete program synthesis.

References (external)

Carried from P22-24's six-cluster citation stack. Add for P25:

- **Mixture of experts** (Shazeer et al.) — disjoint per-role heads via gated routing
- **Causal abstraction identifiability** under partial intervention (Beckers, Halpern)
- **Vervaeke** relevance realization (lecture series) — relevance under representational bounds

Plus the philosophical correlates the program operationalizes: Heidegger, Gibson, Uexküll, enactivism (Maturana/Varela/Thompson/Di Paolo), cybernetics (Ashby), active inference (Friston), pragmatism (Dewey), Canguilhem, Jonas, Simondon.

Pre-registration

`papers/role_specific_identifiability/preregistration.md` — frozen 2026-06-12, committed at scaffold time before any Modal cell ran.

Artifacts

- `artifacts/role_specific_identifiability/sweep_v1.json`